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ORIGINAL ARTICLE

The Contributions of Attentional Bias and Expectancy Bias to Fluctuations in State Anxiety

James Tough[1] · Ben Grafton[1] · Colin MacLeod[1] · Bram Van Bockstaele[1,2]

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Abstract

Background Individuals differ in the degree to which they experience elevations in state anxiety in response to anticipated potentially stressful events, which may be explained by variance in the way information relevant to those stressors is processed. The present study aimed to test the validity of the hypothesis that the attentional processing of negative event-related information will be positively associated with changes in event-related negative expectancy bias, which in turn will be positively associated with elevations in state anxiety.

Method Using a variant of the dual probe task, participants were asked to report their state anxiety and negative expectancy bias at multiple points and were presented with information relevant to a specific anticipated potentially stressful event from which a measure of attentional bias was obtained.

Results The results indicated a case of complete mediation, whereby changes in event-related negative expectancy bias fully mediated the relationship between attentional bias to negative event-related information and elevations in state anxiety.

Conclusions This finding adds to our understanding of how the biased processing of information can explain fluctuations in state anxiety and invites research into alternative presentations of biased information processing.

Keywords Cognition · Emotion · Attention · Expectancy · Anxiety

Elevations in state anxiety are often triggered by exposure to information about an anticipated potentially stressful event, such as talking to friends about upcoming social situations, discussing occupational challenges with colleagues, or researching medical procedures. Importantly, people differ in the degree to which this information serves to elevate their state anxiety. In some individuals, exposure to information about an anticipated potentially stressful event elicits strong elevations of state anxiety, while others instead experience little or no elevation of state anxiety (Rudaizky et al., 2012). Such individual differences in the tendency to experience elevations in state anxiety in response to encountering this information reflects variation in anxiety vulnerability (Spielberger, 1966). As such, it is unsurprising that individuals with heightened anxiety vulnerability display disproportionate increases in state anxiety when encountering information about an anticipated potentially stressful event (Leal et al., 2017). Individuals who report such disproportionate elevations in state anxiety are at greater risk of developing anxiety disorders, and of exhibiting anxiety-linked deficits in social, academic, and cognitive functioning (Mărcuş et al., 2016). Hence, it is important to advance understanding of the mechanisms that underpin individual differences in such elevations in state anxiety resulting from exposure to information about anticipated potentially stressful events.

There is some evidence that individual differences in elevations in state anxiety, observed when people encounter information about an anticipated potentially stressful

✉ James Tough
james.tough@uwa.edu.au

Ben Grafton
ben.grafton@uwa.edu.au

Colin MacLeod
colin.macleod@uwa.edu.au

Bram Van Bockstaele
Bram.VanBockstaele@UGent.be;
Bram.Vanbockstaele@gmail.com

1 Centre for the Advancement of Research on Emotion, School of Psychological Science, University of Western Australia, 35 Stirling Highway, Crawley, WA 6009, Australia

2 Department of Developmental, Personality, and Social Psychology, Ghent University, Henri Dunantlaan 2, 9000 Gent, Belgium

event, may potentially reflect variation in the way they process this information. Specifically, literature suggests that heightened state anxiety responses may be driven by a negative attentional bias (Liu et al., 2019), which involves preferential allocation of attentional resources to negative information relevant to the anticipated potentially stressful event. A common approach to assessing attentional bias utilises the dot probe task (MacLeod et al., 1986). In this task, participants are simultaneously presented with negative and benign information, each presented to different sides of the screen. This information has traditionally been individual emotional words, or static emotional pictures. Following this a small probe is presented on either the left or the right side of the screen and participants are tasked with identifying this probe. It is assumed that participants will be quickest to identify probes that appear in the location they are attending to. Thus, a measure of negative attentional bias is obtained by computing relative speeding to identify probes presented in the locus of negative information, compared to probes presented in the locus of benign information. Many prior studies have established that this index of negative attentional bias is correlated with questionnaire measures of anxiety vulnerability, which assess tendency to have frequently experienced state anxiety elevations in the past. Specifically, people who report more frequent past experiences of elevated state anxiety tend to exhibit greater negative attentional bias (Bar-Haim et al., 2007). As yet, however, researchers have not investigated whether such variation in negative attentional bias to information relevant to an anticipated potentially stressful event predicts variation in the degree to which this information impacts state anxiety. The first purpose of the present study was to determine whether individual differences in the impact of information about the anticipated potentially stressful event on fluctuations in state anxiety varies as a function of how attentional resources are distributed towards that information.

The conventional dot probe measure of negative attentional bias has two limitations that the present experiment sought to circumvent. First, it assesses attentional bias only to individual words or images, which do not carry sufficient informational content to convey detail concerning specific anticipated potentially stressful events. Second, the conventional dot probe task has been subject to criticism concerning its low psychometric reliability (McNally, 2019; Rodebaugh et al., 2016). Fortunately, both limitations have been overcome by a recently developed dual probe task variant that assesses attentional bias to richer and more complex negative and benign information, in a manner that demonstrates good psychometric reliability (Grafton et al., 2021). In this dual probe task, participants are simultaneously presented with two streams of continuous information, one appearing on the right and one on the left side of the screen. During the presentation of these information streams a pair of probes briefly appears, for only 200 ms, with one probe presented in the locus of each information stream. The two probes in each presented pair differ in identity. Participants are required to identify whichever probes they see. It is assumed that they will more often see probes that are presented in the location of the information stream they are attending to. Therefore, an index of attentional bias to negative information is provided by computing the proportion of correctly identified probes that were presented in the locus of the negative information stream. Grafton et al. (2021) not only confirmed that participants who reported having frequently experienced elevated state anxiety in the past displayed greater attention to the negative information streams, but also demonstrated that the index of negative attentional bias provided by this dual probe approach has high psychometric reliability (yielding a Spearman-Brown Coefficient of 0.97). By using this dual probe approach within the present study, we can (a) expose participants to information concerning the anticipated potentially stressful event, and (b) assess the degree to which participants display an attentional bias to negative rather than benign information about this anticipated potentially stressful event. In this way, we will be able to determine whether a greater tendency to preferentially attend to negative information about an anticipated potentially stressful event is characterised by a greater tendency to experience elevations in state anxiety.

It is interesting to consider how such negative attentional bias to event-related information could influence the degree to which this information elevates state anxiety in people. While it is possible that state anxiety may be elevated as a direct result of negatively biased attentional processing of event-relevant information, another possibility is that state anxiety is elevated indirectly through the influence of this negative attentional bias on event-related expectancies. Specifically, individuals who attend more to negative event-relevant information may consequentially develop a negative expectancy bias, reflecting expectancies that are, on average, more negative than benign (Reynolds et al., 2023), which may in-turn result in greater elevations in state anxiety. Negative expectancy bias, which can be assessed by simply having participants report the degree to which they have negative and benign expectancies about future events, has been found to be associated with questionnaire-based measures of anxiety vulnerability that index the frequency with which they have generally experienced elevated state anxiety in the past (Aue & Okon-Singer, 2015; Butler & Mathews, 1987; Reynolds et al., 2023; Steinman et al., 2013). For example, while individuals generally tend to focus more greatly on negative, compared to positive, information that may shape their expectancies (Baumeister et al., 2001), individuals approaching an upcoming test who indicate that they often have become state anxious in the past tend to report a greater negative expectancy bias about this future test (Butler & Mathews, 1987). This finding has been replicated with respect to negative expectancy bias concerning anticipated social situations (Smith & Sarason, 1975), medical procedures (Klages et al., 2004), and holidays (Mura, 2010). However, these studies have not determined whether greater negative expectancy bias concerning a specific anticipated potentially stressful event (a) results from the biased attentional processing of information about that event, or (b) predicts greater elevations in state anxiety. Additionally, in a recent review, Aue and Okon-Singer (2015) reported an association between attentional bias and expectancies, suggesting that visual attention may impact individual expectancies. Although this relationship may be well-established, no studies have considered whether greater elevation in state anxiety, as a result of biased attention, may be accounted for by the relationship between attentional bias and negative expectancy bias.

The present study was designed to test the validity of predictions generated by the hypothesis that greater attentional processing of negative event-relevant information will be associated with a greater increase in negative expectancy bias about this event, which in turn will be associated with a greater elevation in state anxiety. This hypothesis is represented in Fig. 1. To test the predictions generated by the proposed account, this study requires that participants be exposed to negative and benign information relevant to an anticipated potentially stressful event in a manner that permits us (i) to assess their negative attentional bias during exposure to event-relevant information using the dual probe approach, (ii) to assess the degree to which participants' negative expectancy bias about the anticipated potentially stressful event increase following exposure to event-relevant information, and (iii) to assess the degree to which participants' state anxiety becomes elevated following exposure to event-relevant information.

The hypothesis under test generates the following four experimental predictions: (i) higher scores on the measure of negative attentional bias will be associated with higher scores on the measure of state anxiety elevation; (ii) higher scores on the measure of change in negative expectancy bias will be associated with higher scores on the measure of state anxiety elevation; (iii) higher scores on the measure of negative attentional bias will be associated with higher scores on the measure of change in negative expectancy bias; and (iv) the association between the negative attentional bias scores and the state anxiety elevation scores will be mediated by change in negative expectancy bias scores.

Transparency and Openness

In the following, we report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. Information and resources relevant to the development of this study and data analysis are stored on the UWA Institutional Research Data Storage (IRDS). This study was funded by an Australian Research Council Discovery Early Career Research Award presented to Ben Grafton.

Method

The experimental design required that participants believed they would experience a potentially anxiety-eliciting event at the end of the test session. As such, participants were told that at the end of the session they would watch a film concerning the experience of refugees, which could potentially be distressing because it would convey hardship and suffering, but that could potentially be uplifting because it would show how the human spirit can triumph over adversity. This putative film viewing experience (which did not actually occur) will be referred to throughout the following description of methodology as the 'anticipated potentially stressful event'.

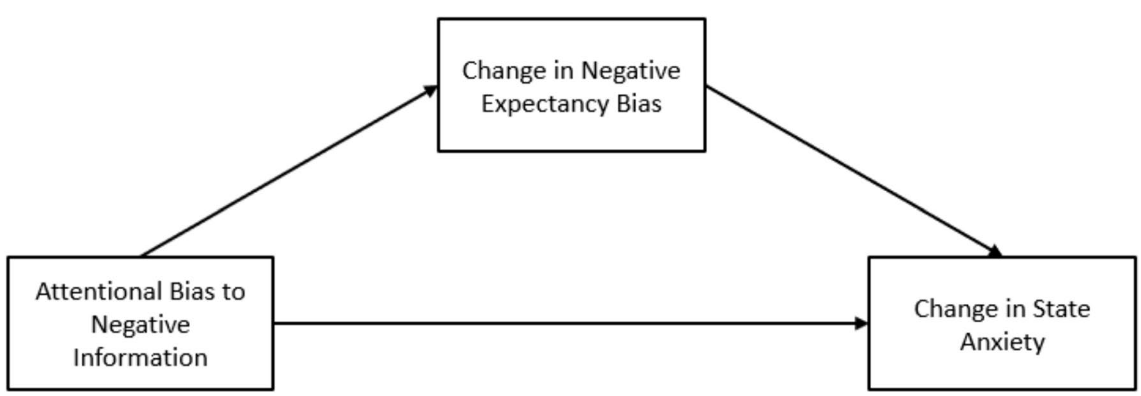

Fig. 1 Model representing the hypothesized relationship between negative attentional bias, change in negative expectancy bias, and change in state anxiety

Participants

The participants in this study were obtained from a community sample of adults. As such, one hundred forty-one participants (84 self-identified as male; remainder self-identified as female) were recruited via Amazon Mechanical Turk (2017) and received monetary compensation upon completion of the experiment. The mean age of participants was 39.37 years ($SD = 10.75$; range = 19–65). All participants resided in the United States of America.

Questionnaires

Negative Expectancy Bias Assessment

This study required that we assess the degree to which participants displayed negative expectancy bias, reflecting relatively inflated negative expectancies compared to positive expectancies about the anticipated potentially stressful event. To assess negative expectancy bias, we presented participants with an 8-item instrument designed to assess their emotional expectancies concerning the film viewing experience. Half of these items were Positive Expectancy Items, which described an emotion that represents a generally positive experience, such as 'reassured' or 'uplifted'. The other half of these items were Negative Expectancy Items, which described an emotion that represents a generally negative experience, such as 'appalled' or 'saddened'. Participants were asked to rate the degree to which they expected to experience each Positive Expectancy Item and Negative Expectancy Item while viewing the film. This assessment scale was computer delivered. Each Positive Expectancy Item and Negative Expectancy Item was displayed in the centre of the screen, below which a sliding Likert scale that ranged from 1 to 100 was presented, respectively labelled 'not at all' and 'very much'. Participants were required to move the marker to the point on this scale which reflected how strongly they expected to feel that emotion specified in each item after viewing the anticipated refugee film. Scores on each of these items could range from 1 (low expectancy) to 100 (high expectancy). A Negative Expectancy Bias Score was computed by subtracting the participant's average score for the Positive Expectancy Items from their average expectancy score for the Negative Expectancy Items, resulting in a total score that represents the degree to which more negative expectancies, compared to positive expectancies, were endorsed. As such, a positive score indicates that, on average, expectancies were more negative, while a negative score indicates that, on average, expectancies were more positive. A score of zero indicates an approximately equal proportion of negative and positive expectancies. The internal consistency reliability of this measure across two assessment points (see procedure) was good, Cronbach's $\alpha = 0.79$ and 0.74, respectively.

State Anxiety Assessment

This study also required that we assess the degree to which participants experienced state anxiety during the experiment. We assessed state anxiety using the short-form of the Spielberger State-Trait Anxiety Inventory (STAI-6; Marteau & Bekker, 1992), which is a well-established measure of state anxiety that has high discriminant and convergent validity and internal consistency reliability (Marteau & Bekker, 1992). While there are other well-validated measures of state-anxiety, such as the STAI-5 (Zsido et al., 2020), the STAI-6 was selected due to its established sensitivity to fluctuations in state anxiety (Marteau & Bekker, 1992). In the STAI-6, respondents are presented with six items each pertaining to a symptom of state anxiety. Three of these items are positively associated with high state anxiety, such as 'I feel tense', and three are negatively associated with high state anxiety, such as 'I feel calm'. Participants are asked to indicate the degree to which each item describes their current emotional state using a 100-point Likert scale, with possible scores ranging from 1 (not at all) to 100 (very much). State Anxiety Scores were calculated as the mean score across the six items after reverse scoring the items negatively associated with high state anxiety.

Information Statements Describing Stressor

It was a requirement of this study that participants would be presented with both emotionally negative and benign information about the anticipated potentially stressful event. To meet this requirement, two sets of 64 statements were created, respectively providing negative and benign information supposedly conveyed by former viewers of the refugee film. Each of the 64 Negative Information Statements described both a negative content component of the film and the supposed viewer's negative emotional reaction to that content (e.g., "watching the boy return home to find his stuffed bear charred and matted was very demoralising"). Each of the 64 Benign Information Statements described both a benign content component of the film and the supposed viewer's benign emotional reaction to that content (e.g., "watching the siblings laugh and tease each other over dinner was an enlivening experience"). Each statement ranged from 90 to 115 characters in length, and no two statements described the same film content component or used the same adjective used to describe the emotional reaction content.

To ensure that participants understood the information presented to them, two comprehension questions were created for each of the Negative Information Statements and Benign Information Statements. Each Negative Information

Statement Comprehension Question pair and Benign Information Statement Comprehension Question pair contained questions that referred to the event described in the statement (e.g., "Was the daughter separated from her family?") or questions that referred to the emotional response it elicited in the person who wrote the statement (e.g., "Was it enraging to see how the children were treated?"), with equal frequency. To permit the balancing of correct responses, each pair of comprehension questions contained a question where the correct response was 'yes' and a question where the correct response was 'no'. Participants who answered comprehension questions with an accuracy rate of less than 75% were excluded from analysis on the grounds of not having adequately understood the presented information.

The 128 statements used in this study were validated by a cohort of independent raters who were asked to indicate, for each statement, how negative or benign they would expect to feel when watching the refugee video if all they had seen was that statement. These individuals indicated their answer by adjusting a slider that ranged from 0 to 100, where a score of 0 indicated that they expected to feel negative when watching the refugee video and a score of 100 indicated that they expected to feel benign. A paired-samples t-test revealed a significant difference between the average negativity ratings of the Benign Information Statements ($M = 77.50$; $SD = 15.0$) compared to the Negative Information Statements ($M = 22.5$; $SD = 22.6$), $t(18) = 6.87$, $p < 0.001$, as required.

Information Statement Presentation and Attentional Bias Assessment

To present participants with Negative Information Statements and Benign Information Statements and assess attentional bias to Negative Information Statements, a variant of the dual-probe task (Grafton et al., 2021) was employed. In each of the 64 trials of this task, participants were presented with one of the 64 Negative Information Statements and one of the 64 Benign Information Statements for 6000 ms,[1] during which time a pair of probes was briefly presented to assess relative attention to the two statements (2000 ms after the statements are presented). Participants were then required to respond to either the Negative Information Statement Comprehension Question or the Benign Information Statement Comprehension Question.

At the beginning of the trial, participants were informed as to the location of the Negative Information Statement and Benign Information Statement, as indicated by a pair of blue and yellow squares with a plus or minus symbol inside them which appeared for 2000 ms. Each coloured square corresponded to the valence of the information that would be presented in that location. For half the participants the blue and yellow square respectively indicated the position of the Benign Information Statement and the Negative Information Statement, as in Fig. 2 where the blue square contains a '+' symbol and the yellow square contains a '−' symbol, whereas for the other half the opposite was the case. Following this and prior to the presentation of the Negative Information Statement and Benign Information Statement, participants were reminded of the location in which these statements would appear by the coloured squares which briefly appeared again inside the white box. Each statement then appeared in one of two white boxes, which were either horizontally aligned (one on the left half of the screen and one on the right half of the screen) or vertically aligned (one on the top half of the screen and one on the bottom half of the screen). The frequency with which each of these conditions occurred was counterbalanced across participants, such that each participant was exposed to an equal number of horizontally and vertically aligned white boxes throughout the task. This is to ensure participants' attention was captured by the stimulus rather than a particular location on the screen. During the presentation of the Negative Information Statement and Benign Information Statement, a pair of probes, which appeared in the location of the two statements, appeared for 200 ms. These probes were small three-by-three grids in which one of the outer eight positions was shaded in grey. The probes within each pair were always different, but participants were simply told to indicate the identity of whichever probe they saw by pressing the key on the three-by-three number pad which corresponded to the position of the grey shaded square in that probe. Following presentation of the two Information Statements, two Information Statement Comprehension Questions were then displayed. One of these questions was related to the Negative Information Statement, and the other to the Benign Information Statement, that had just been presented, with each appearing in the same location previously occupied by the corresponding Information Statement. Participants were required to answer the comprehension question that corresponded to the Information Statement they had attended to on that trial, with either a 'yes' ('y' key) or 'no' ('n' key). A graphical illustration of a single trial of this task is displayed in Fig. 2.

Following the approach adopted by Grafton et al. (2021) the index of Negative Attentional Bias was computed by expressing the number of trials on which participants correctly identified the probe in the locus of the Negative Information Statement as a proportion of the total number of trials on which either probe was correctly identified. Thus, the

[1] Note that the duration of the information presentation was determined based on the average reading speed of 20 characters per second (Brysbaert et al., 2021). The average number of characters in the statements used in the present study was 105; as such, a duration of 6 s was determined as sufficient to ensure the statements were read but not rehearsed.

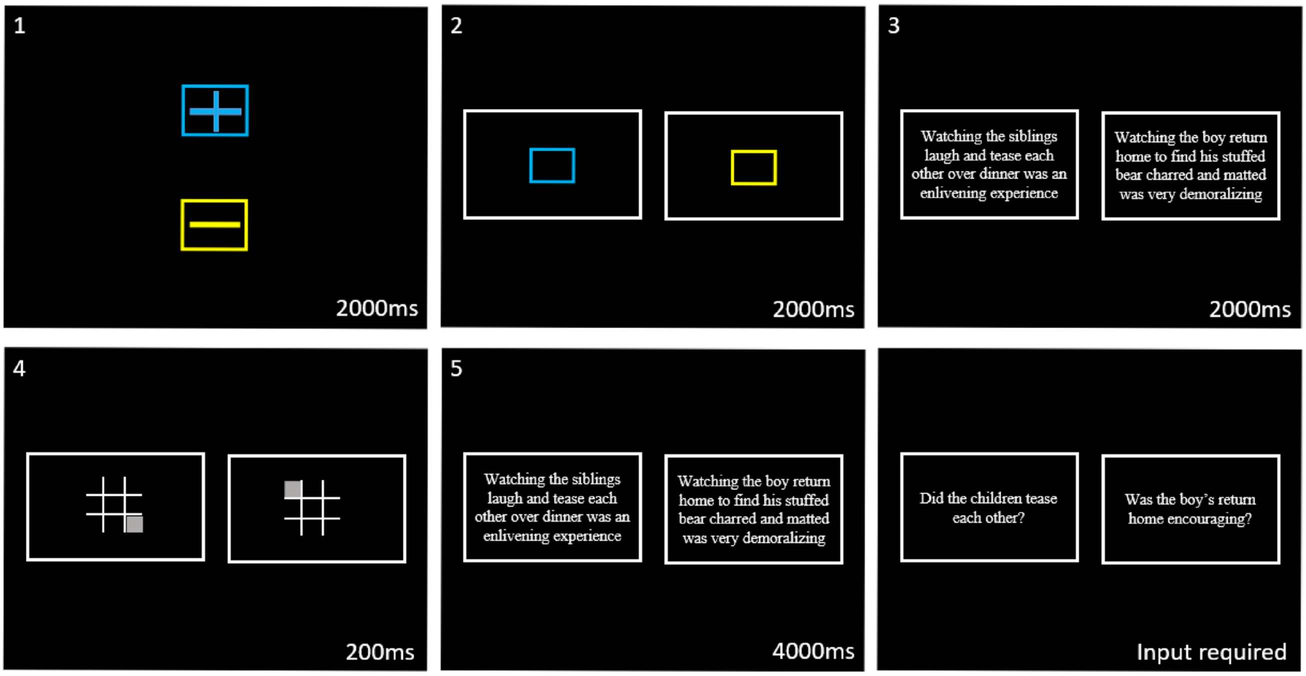

Fig. 2 Diagram depicting the sequence of events for a single trial of the Information Statement Presentation and Attentional Bias Assessment

Negative Attentional Bias index resulted from the following formula:

$$\text{Negative Attentional Bias index} = \text{Number of probes correctly identified in locus of Negative Information Statement} / \text{Total number of probes correctly identified}$$

Scores on this measure could range from values of 0 to 1, where a score of 0 would indicate maximal attentional avoidance of the Negative Information Statements and a score of 1 would indicate maximum attentional vigilance to Negative Information Statements. The internal consistency reliability of this measure was computed, confirming that the dual-probe task is a highly reliable measure of attentional bias, Cronbach's $\alpha = 0.97$.

Software & Hardware

The attentional assessment task and questionnaires were delivered using Inquisit by Millisecond (Inquisit, 2016), version 5.0.14.0. Participants completed the task using their own personal computers.

Procedure

Participants first provided informed consent, then provided demographic information. Following this, participants were told that at the end of the experiment they would be watching a refugee film and were informed that this film could potentially be distressing by conveying hardship and suffering, but also could be uplifting as it would show how the human spirit can triumph over adversity. Participants were then asked to complete the State Anxiety Assessment and Negative Expectancy Bias Assessment, so that Pre-Information Statement Presentation measures of state anxiety and negative expectancy bias could be obtained. They were then provided with instructions for the Information Statement Presentation and Attentional Bias Assessment, after which they completed a practice block of the task in which neutral statements were used. Participants then completed the Information Statement Presentation and Attentional Bias Assessment component, which resulted in the measure of Negative Attentional Bias. Following completion of this task, participants were asked to complete the State Anxiety Assessment and Negative Expectancy Bias Assessment again, so that Post-Information Statement Presentation measures of state anxiety and negative expectancy bias could be obtained. An index of Change in Negative Expectancy Bias was computed as the degree to which negative expectancy bias scores were greater at the Post-Information Statement Presentation assessment compared to the Pre-Information Statement Presentation assessment. An index of State Anxiety Elevation was computed as the degree to which state anxiety scores

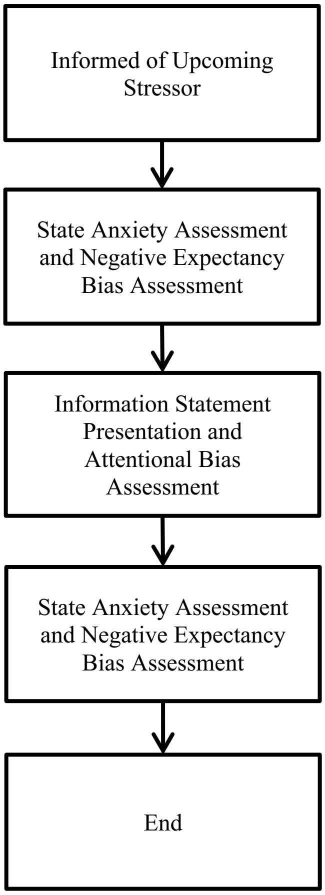

Fig. 3 Diagram depicting the sequence of events during the experiment

were greater at the Post-Information Statement Presentation assessment compared to the Pre-Information Statement Presentation assessment. The experiment then ended, with participants being debriefed and thanked for their participation. This sequence of events is illustrated in Fig. 3.

Results

Data Cleaning and Preparation

Of the 141 participants, 17 failed to meet the comprehension criteria and so were eliminated from the analyses. An additional 2 participants were removed due to outlying scores, with one participant identified as an outlier on the measure of change in state anxiety and the other identified as an outlier on the measure of change in negative expectancy bias. Outliers were identified as scores that fell outside three times the inter-quartile range from either the lower or upper quartile, as per the guidelines recommended by Field et al. (2012). As such, the final sample comprised 122 participants.

Descriptive Statistics

A summary of the descriptive statistics is outlined in Table 1. A paired-samples t-test revealed a significant increase in state anxiety scores after ($M = 33.90$, $SD = 22.60$), compared to before ($M = 25.70$, $SD = 19.30$), the Information Statement Presentation and Attentional Bias Assessment, $t(121) = 5.37$, $p < 0.001$. Another paired-samples t-test revealed a significant reduction in negative expectancy bias after ($M = -0.46$, $SD = 35.50$), compared to before ($M = 10.40$, $SD = 28.40$), the Information Statement Presentation and Attentional Bias Assessment, $t(121) = -3.59$, $p < 0.001$. Additionally, the proportion of dual probe presentations on which a probe was correctly identified was calculated as 85.65% ($SD = 10.12\%$).

Did Negative Attentional Bias Predict State Anxiety Elevation?

To test the veracity of the prediction generated by the hypothesis, that higher scores on the measure of negative attentional bias will be associated with higher scores on the measure of state anxiety elevation, a simple linear regression analysis was conducted. In this analysis, the measure of state anxiety elevation was entered as the dependent variable and the measure of negative attentional bias was entered as the predictor variable.[2] The results of the regression indicated that negative attentional bias scores explained 9.75% ($R^2 = 0.0975$) of the variation in state anxiety elevation, $F(1, 120) = 13.0$, $p < 0.001$. In this model there was also a significant, positive association between the measure of negative attentional bias and the measure of state anxiety elevation, $\beta = 0.312$, $p < 0.001$. This analysis satisfied the assumptions of autocorrelation (Durbin–Watson Test = 1.89, $p = 0.516$) and heteroskedasticity (Breusch–Pagan Test = 0.196, $p = 0.658$). However, the assumption of normality of errors was violated (Kolmogorow–Smirnov Test = 0.071, $p = 0.566$). This analysis was repeated using cubic root transformed data to account for this with no change to the significance of the results. Given the transformation of dependent variables can impact the interpretability of the reported effects, the untransformed variables are reported above.

[2] This analysis was also completed using a model where the measure of state anxiety after the task was entered as the dependent variable, the measure of state anxiety before the task was entered as a predictor in the first block, and the measure of negative attentional bias was entered as a predictor variable in the second block. This analysis allowed the testing of the unique variance in state anxiety after the task accounted for by negative attentional bias, controlling for state

Table 1 Descriptive statistics for negative attentional bias, negative expectancy bias (pre- and post-task), and state anxiety (pre- and post-task)

	N	Mean	Median	SD	Minimum	Maximum
State anxiety pre-task	122	25.70	23.10	19.30	1.00	90.00
State anxiety post-task	122	33.90	31.80	22.60	1.00	87.30
Negative expectancy bias pre-task	122	10.40	7.75	28.40	−63.00	74.00
Negative expectancy bias post-task	122	−0.46	−2.13	35.50	−71.30	69.00
Negative attentional bias	122	0.26	0.16	0.28	0.00	1.00

Did Change in Negative Expectancy Bias Predict State Anxiety Elevation?

To test the veracity of the prediction generated by the proposed hypothesis, that higher scores on the measure of change in negative expectancy bias will be associated with higher scores on the measure of state anxiety elevation, a simple linear regression analysis was conducted. In this analysis, the measure of state anxiety elevation was entered as the dependent variable and the measure of change in negative expectancy bias was entered as the predictor variable.[3] The result of the analysis indicated that change in negative expectancy bias scores explained 22.6% ($R^2 = 0.226$) of the variance in state anxiety elevation, $F(1, 120) = 35.0$, $p < 0.001$. In this model there was also a significant, positive association between the measure of change in negative expectancy bias and the measure of state anxiety elevation, $\beta = 0.475$, $p < 0.001$. This analysis satisfied the assumptions of normality of errors (Kolmogorov–Smirnov Test = 0.119, $p = 0.063$), autocorrelation (Durbin–Watson Test = 1.92, $p = 0.672$), and heteroskedasticity (Breusch–Pagan Test = 3.26, $p = 0.071$).

Did Negative Attentional Bias Predict Change in Negative Expectancy Bias?

To test the veracity of the prediction generated by the hypothesis, that higher scores on the measure of negative attentional bias will be associated with higher scores on the measure of change in negative expectancy bias, a simple linear regression analysis was conducted. In this analysis, the measure of change in negative expectancy bias was entered as the dependent variable and the measure of negative attentional bias was entered as the predictor variable.[4] The results of the regression indicated that negative attentional bias scores explained 29.2% ($R^2 = 0.292$) of the variation in change in negative expectancy bias, $F(1, 120) = 49.4$ $p < 0.001$. In this model there was also a significant, positive association between the measure of negative attentional bias and the measure of change in negative expectancy bias, $\beta = 0.540$, $p < 0.001$. This analysis satisfied the assumptions of normality of errors (Kolmogorov–Smirnov Test = 0.071, $p = 0.566$), autocorrelation (Durbin-Watson Test = 1.89, $p = 0.516$), and heteroskedasticity (Breusch-Pagan Test = 0.196, $p = 0.658$).

Figures illustrating the associations outlined above are included in the Appendix (see Figs. 4, 5, and 6).

Was the Prediction of State Anxiety Elevation by Negative Attentional Bias Mediated by Change in Negative Expectancy Bias?

To test the validity of the prediction that the association between negative attentional bias scores and state anxiety elevation scores will be mediated by change in negative expectancy bias scores, a mediation analysis was conducted. This was accomplished via the PROCESS macro for SPSS (model 4) developed by Hayes (2013), where Negative Attentional Bias scores were entered as the predictor variable, the index of State Anxiety Elevation was entered as the outcome variable, and the index of Change in Negative Expectancy Bias was entered as the mediator variable.

Variance in Negative Attentional Bias scores did statistically predict variance in the index of Change in Negative

Footnote 2 (continued)

anxiety before the task. The pattern and significance of the results were unchanged, so the original analysis was reported.

[3] This analysis was also completed using a model where the measure of state anxiety after the task was entered as the dependent variable, the measures of state anxiety and negative expectancy bias before the task were entered as predictors in the first block, and the measure of negative expectancy bias after the task was entered as a predictor in the second block. This analysis allowed for the measurement of the unique variance in state anxiety after the task accounted for by negative expectancy bias after the task, controlling for both state anxiety and negative expectancy bias before the task. The pattern and significance of the results were unchanged, so the original analysis was reported.

[4] This analysis was also completed using a model where the measure of negative expectancy bias after the task was entered as the dependent variable, the measure of negative expectancy bias before the task was entered as a predictor in the first block, and the measure of negative attentional bias was entered as a predictor variable in the second block. This analysis allowed the testing of the unique variance in negative expectancy bias after the task accounted for by negative attentional bias, controlling for negative expectancy bias before the task. The pattern and significance of the results were unchanged, so the original analysis was reported.

Expectancy Bias (a path; $\beta = 0.54$, $p < 0.001$), and variance in the index of Change in Negative Expectancy Bias did statistically predict variance in the index of State Anxiety Elevation (b path; $\beta = 0.43$, $p < 0.001$). Of most relevance to the hypothesis under scrutiny is the finding that variance in Negative Attentional Bias scores did statistically predict variance in the index of State Anxiety Elevation in a manner that was mediated by variance in the index of Change in Negative Expectancy Bias (ab path; $\beta = 0.23$). The significance of this indirect effect was tested using bootstrapping procedures, where standardized indirect effects were computed for each of the 5000 samples and the 95% confidence intervals were computed by determining the indirect effects at the 2.5th and 97.5th percentiles, 95% CI [0.108, 0.375]. As the confidence intervals did not include zero, the indirect effect was statistically significant. This model reflects a case of complete mediation of the relationship between Negative Attentional Bias scores and the index of State Anxiety Elevation by the index of Change in Negative Expectancy Bias, as controlling for the effect of the index of Change in Negative Expectancy Bias reduced the total effect of Negative Attentional Bias scores on the index of State Anxiety Elevation (c path; $\beta = 0.31$, $p < 0.001$) to non-significance (c' path; $\beta = 0.08$, $p = 0.494$). As such, these results suggest that greater Negative Attentional Bias scores indirectly predict a heightened index of State Anxiety Elevation through its impact on the index of Change in Negative Expectancy Bias.[5] A Monte Carlo power analysis for mediation analyses was conducted using a simulation-based method developed by Qin (2024) to determine the power associated with an established sample size. At a significance criterion of a = 0.05 and a sample size of N = 122, the serial mediation effect was detected with 97% power across a simulation with 5000 replications.

[5] An alternate pathway, considering whether a heightened index of Change in Negative Expectancy Bias indirectly predicts a heightened index of State Anxiety Elevation through the mediating impact of Negative Attentional Bias, was tested using the same mediation model and parameters as reported above. In this model, the index of Change in Negative Expectancy Bias scores was entered as the predictor variable, the index of State Anxiety Elevation was entered as the outcome variable, and Negative Attentional Bias scores were entered as the mediator variable. The results returned a non-significant indirect effect, suggesting that the relationship between the index of Change in Negative Expectancy Bias and the index of State Anxiety Elevation is not mediated by Negative Attentional Bias to event-relevant information.

Discussion

The present study aimed to test the validity of predictions generated by the hypothesis that the attentional processing of event-relevant information will be positively associated with change in negative expectancy bias about this event, which will in turn be positively associated with changes in state anxiety. In the following, we will discuss how the results obtained bear upon the predictions generated by this hypothesis and consider the theoretical and practical implications of these findings. Some limitations of the current study then will be acknowledged before potential future research directions are suggested.

The results firstly revealed, as predicted by the hypothesis under scrutiny, that individuals who demonstrate greater attentional bias to negative information, compared to benign information, about the anticipated potentially stressful event are more likely to also experience a greater elevation in their state anxiety. This indicates that the degree to which individuals experience elevations in state anxiety about the anticipated potentially stressful event is positively associated with the degree to which they attend to negative, compared to positive, event-relevant information. Whilst a plethora of research has established that negative attentional bias shares an association with questionnaire-based measures of the frequency with which state anxiety has previously been experienced (Bar-Haim et al., 2007), this finding highlights the association between negative attentional bias and in-vivo elevations in state anxiety.

The results also revealed that, as predicted by the hypothesis, individuals who report an increasingly disproportionate endorsement of negative expectancies, compared to positive expectancies, over time, were also more likely to experience greater elevations in state anxiety. Whilst previous studies have established that negative expectancy bias scores correlate with questionnaire-based measures of how frequently state anxiety has been experienced in the past (Butler & Mathews, 1987; Cabeleira et al., 2014; Klages et al., 2004; Smith & Sarason, 1975), the present study utilised a dynamic measure of state anxiety to expand this knowledge in two ways: the results of the present study pertain to (i) fluctuations in state anxiety during the experiment, (ii) resulting from changes in negative expectancy bias about the event for which both state anxiety and negative expectancy bias were assessed. The benefit of this approach to assessing fluctuations in state anxiety comes from the resulting capacity to associate the elevation of state anxiety with (a) the processing of information relevant to this specific event during the experiment, and (b) to the change in negative expectancy bias about this specific event. We suggest that future studies designed to advance understanding of the relationship between fluctuations in state anxiety and cognitive biases implicated in the processing of information about anticipated stressors could potentially benefit from adopting this methodological approach.

As predicted by the hypothesis, the results also revealed that individuals who report a greater negative attentional bias are also more likely to experience an increasingly negative expectancy bias over time. While this finding concurs with previous studies which have found that variance in visual attention to threatening information predicts variance in negative expectancy bias (Aue & Okon-Singer, 2015; Aue et al., 2013a), it also highlights that biased event-relevant information processing is associated with changes in negative expectancy bias. This was achieved by incorporating a dynamic measure of negative expectancy bias which permits an assessment of how it changes in relation to the biased processing of event-relevant information. Although this finding contributes to our understanding of the relationship between negative attentional bias and changes in negative expectancy bias, future studies may wish to lend consideration to the potential variation in the strength of negative expectancies, compared to positive expectancies, when assessing negative expectancy bias. Previous research has suggested that negative information carries greater emotional weight positive information, which may impact decision making (Muller-Pinzler et al., 2019). Future studies may be able to address this natural variation in the emotional strength of valenced information by establishing a strength threshold such that expectancies of similar strengths can be used to generate a negative expectancy bias score. Additionally, while the present study found no evidence that individuals' pre-existing expectancies impact the degree to which they preferentially attend to negative information, it is possible that under certain circumstances pre-existing expectancies resulting from past emotional experiences may drive negative attentional bias towards potential threats, such as in phobic individuals (Aue et al., 2013a, 2013b).

Of critical importance to the hypothesis under test, we found that variation in change in negative expectancy bias fully mediated the relationship between negative attentional bias to event-relevant information and state anxiety elevation. Thus, the relationship between negative attentional bias and state anxiety elevation reflected an indirect association, whereby the positive association between negative attentional bias and state anxiety elevation was fully explained by their positive associations with change in negative expectancy bias. This observed association is critically informative to our understanding of the operation of these biases, in that it suggests that, when provided with equal opportunity to process information relevant to an anticipated potentially stressful event, biased attention towards negative information is associated with the endorsement of increasingly disproportionate negative, compared to positive, expectancies. This finding significantly advances our understanding of the mechanisms that may contribute to fluctuations in state anxiety. However, it should be recognised that the obtained mediation effect does not itself permit causal conclusions. Thus, we suggest that future researchers more directly investigate the causal nature of the currently observed pattern of associations. This could be achieved by systematically manipulating attentional bias to information relevant to an anticipated stressor, using established attentional bias modification procedures (Mogg et al., 2017), to determine whether such manipulation has a subsequent impact on negative expectancy bias, and in turn elevations in state anxiety. If so, this would suggest that interventions designed to attenuate negative attentional bias could be of potential therapeutic value to individuals who tend to experience problematic degrees of state anxiety elevation after being exposed to information relevant to an anticipated stressor.

It is important to recognize some limitations of the present study. This study permits conclusions only about how the processing of an acute package of information about an anticipated stressor serves to acutely impact on state anxiety across a forty-minute time-period. Whilst fluctuations in state anxiety may certainly occur over brief periods of time, even as short as forty-five seconds (Grillon et al., 1993), after processing information about anticipated stressors, it is common for people to also experience much longer periods where state anxiety fluctuates in response to prolonged, sporadic processing of information relevant to that stressor, such as pregnancy (Schetter, 2011) and examinations (Butler & Mathews, 1987). As such, future studies should consider implementing a methodology where participants are subject to multiple, or protracted, exposures to information relevant to an anticipated stressor to ascertain whether attentional bias is associated with state anxiety elevation in a way that is mediated by change in negative expectancy bias across this extended period. Another potential limitation of the present study is that all data was collected remotely via an online platform. Whilst there are benefits associated with this approach, such as having access to large heterogeneous samples which provide greater statistical power and reduce experimental costs (Birnbaum, 2004; Reips, 2000), there are several drawbacks associated with online testing. Primarily, the lack of experimental control poses potential issues for the validity of the data, specifically concerning the liberty with which extraneous variables can vary (Reips, 2000), which may impact the generalisability of the findings. It is therefore recommended that future studies test the robustness of the present findings by replicating it under lab conditions. An additional benefit of any such lab-based replication is the opportunity to exploit eye-tracking technologies within the attentional assessment procedure (Skinner et al., 2018), which could enable future researchers to seek converging evidence for the present results. Finally, we did not assess participants' racial/ethnic identification, socioeconomic status, nor their preconceptions or personal experiences concerning immigration, as the present research question did not necessitate testing the contributions of these

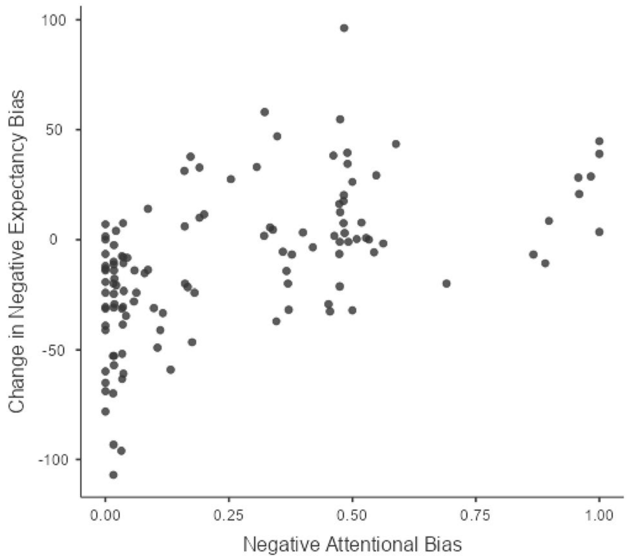

Fig. 4 Scatterplot illustrating association between Negative Attentional Bias scores and Negative Expectancy Bias Residual scores

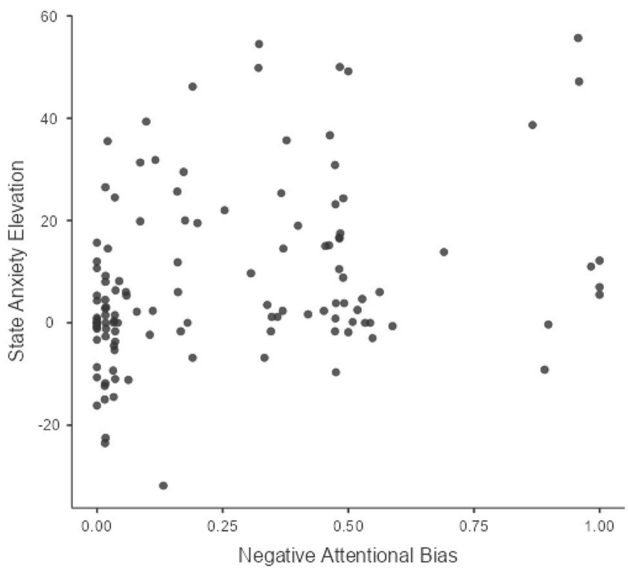

Fig. 5 Scatterplot illustrating association between Negative Attentional Bias scores and State Anxiety Residual scores

characteristics to the relationship between negative attentional bias to event-relevant information, change in negative expectancy bias, or state anxiety elevation. However, future researchers could assess these variables in order to test alternative variants of the current hypothesis that implicate them.

To conclude, the present study used the dual-probe task to test the hypothesis that attentional bias to negative information about an anticipated future event drives fluctuations in state anxiety, evidenced by elevations state anxiety after having processed event-relevant information, in a manner that in mediated by change in negative expectancy bias about that event.

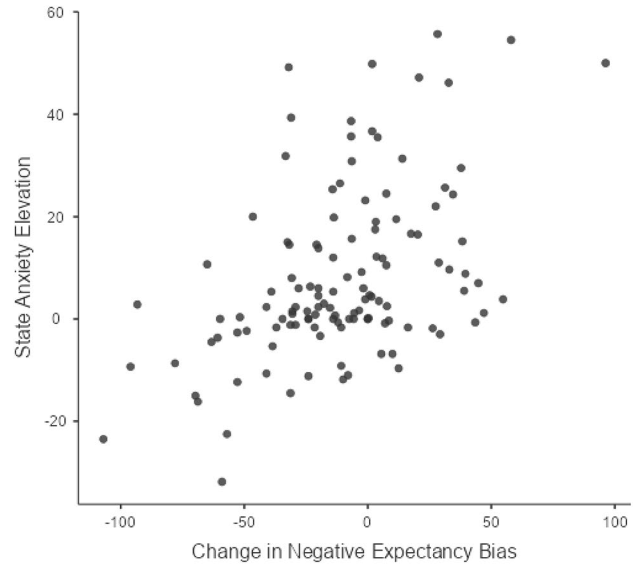

Fig. 6 Scatterplot illustrating association between Negative Expectancy Bias Residual scores and State Anxiety Residual scores

The results of the study support this hypothesis and extend our understanding of the interplay between negative attentional bias, change in negative expectancy bias, and state anxiety elevation, in response to a specific anticipated stressor. We hope these findings will be of interest to fellow researchers investigating the basis of emotional vulnerability, and that adaptations of the novel methodology introduced in this study will be of value to colleagues working on this important topic.

Appendix

Scatterplots Illustrating Correlations

Below are scatterplots illustrating correlations between Negative Attentional Bias scores, Negative Expectancy Bias scores, and State Anxiety scores. Both Negative Expectancy Bias and State Anxiety scores are reported here as residual values, such that the influence of initial expectancies and state anxiety is controlled for (see Figs. 4, 5, 6).

Information Statements Describing Stressor and Comprehension Questions

A list of the Negative Information Statements, Benign Information Statements, and comprehension questions used in the Information Statement Presentation and Attentional Bias Assessment are reported here (see Tables 2 and 3).

Table 2 A list of Benign Information Statements and Benign Information Statement Comprehension Question pairs

	Benign information statements	Benign information statement comprehension questions set 1	Correct answer	Benign information statement comprehension questions set 2	Correct answer
1	It was wonderful to behold their love when the mother was finally reunited with her daughter	Was the mother reunited with her daughter?	YES	Was the mother kept away from her daughter?	NO
2	It was a truly beautiful sight to see the children playing and laughing together so gleefully	Were the children eating together?	NO	Were the children playing and laughing together?	YES
3	It was really inspiring that the father was able to steadfastly maintain such a positive attitude	Was the father's attitude inspiring?	YES	Was the father's attitude uninspiring?	NO
4	It was a profoundly heart-warming gesture when the family shared their rations with the other refugees	Was the family's gesture offensive?	NO	Was the family's gesture heart-warming?	YES
5	It was hugely impressive that the children in the family took such loving care of one another	Did the family take care of one another?	YES	Did the family ignore one another?	NO
6	It was a splendid testimony to human kindness that they helped that lost dog, despite their own difficulties	Did the family leave the lost dog?	NO	Did the family help the lost dog?	YES
7	It was deeply rewarding to see the family’s immense joy when they were welcomed into their new home	Was the family's joy rewarding?	YES	Was the family's joy unrewarding?	NO
8	It was empowering to witness the parents’ strength and fortitude as they worked together to nurture their children	Were the parents' efforts disappointing?	NO	Were the parents' efforts empowering?	YES
9	It was nothing short of charming to see that the children were so steadfastly proud of their parents	Were the children proud of their parents?	YES	Were the children disappointed in their parents?	NO
10	It was breathtaking to witness the unbridled happiness on the faces of the people as they danced	Were the people singing?	NO	Were the people dancing?	YES
11	It was very gratifying to see the family’s generosity repaid by the bus driver who brought them to safety	Was the bus driver's generosity gratifying?	YES	Was the bus driver's generosity ungratifying?	NO
12	It was a fantastic experience to see the old friends meeting together again after so much time apart	Was the reunion underwhelming?	NO	Was the reunion a fantastic experience?	YES
13	It was magnificent that the family showed such consistent compassion, and never failed to help others in need	Did the family always show compassion?	YES	Did the family lack compassion?	NO
14	It was extremely moving when the kind villagers brought the family new shoes and blankets	Did the villagers bring the family food?	NO	Did the villagers bring the family shoes and blankets?	YES
15	It was greatly satisfying to see that the kids are now thriving, with no negative aftereffects of their ordeal	Was the change in circumstances satisfying?	YES	Was the change in circumstances unsatisfying?	NO

Table 2 (continued)

	Benign information statements	Benign information statement comprehension questions set 1	Correct answer	Benign information statement comprehension questions set 2	Correct answer
16	It was uplifting to learn about the work carried out by the people who run the charity that stepped in to help	Was learning about the charity work disheartening?	NO	Was learning about the charity work uplifting?	YES
17	It was extremely pleasing to see the father bring his family a hot meal for the first time in months	Did the family have a hot meal?	YES	Did the family have a cold meal?	NO
18	It was a vivid example of the kindness of others when seeing the family's relief at being spared some bread	Was the family spared some water?	NO	Was the family spared some bread?	YES
19	It was remarkable to behold the joy on everyone's face when the refugee camp was supplied with fresh food	Was the peoples' joy remarkable?	YES	Was the peoples' joy unremarkable?	NO
20	It encapsulated humanity's capacity for gratitude when the children graciously accepted food offerings	Was the childrens' gratitude lacking?	NO	Was the childrens' gratitude satisfying?	YES
21	It was a stirring moment to witness the willingness of the shoemaker to freely give the family new footwear	Did the family receive new footwear?	YES	Did the family receive new shirts?	NO
22	It was incredibly inspiriting to see the compassion displayed by the doctors when providiing free treatment	Did the doctors charge people for treatment?	NO	Did the doctors provide free treatment?	YES
23	It was a moment of undeniable relief when the sick child was provided with medication to enable his recovery	Was the boy's recovery relieving?	YES	Was the boy's recovery alarming?	NO
24	It was a true testament to human compassion to see the generosity of the neighbours when the wife went into labour	Did the neighbours turn a blind eye to the wife in labour?	NO	Did the neighbours help the wife in labour?	YES
25	It was truly commendable how the father showed respect towards the strangers giving up their medical supplies	Did the strangers give away their medical supplies?	YES	Did the strangers refuse to part with their medical supplies?	NO
26	It was a striking spectacle to behold the indisputable joy displayed by the brothers when reunited at the camp	Were the sisters reunited at the camp?	NO	Were the brothers reunited at the camp?	YES
27	It was very sweet to see the husband and wife share a moment of peaceful affection towards one another	Was the moment between the husband and wife sweet?	YES	Was the moment between the husband and wife emotionless?	NO
28	It was delightful seeing the smiles on the faces of the children when playing ball with the other kids	Were the children's faces a dispiriting sight?	NO	Were the children's faces a delightful sight?	YES
29	It was an enlivening experience watching the siblings laugh and tease each other over dinner	Did the children tease each other?	YES	Did the children shout at each other?	NO

Table 2 (continued)

	Benign information statements	Benign information statement comprehension questions set 1	Correct answer	Benign information statement comprehension questions set 2	Correct answer
30	It was a great source of fulfillment seeing the family's visible relief when welcomed into their new home	Was the family underwhelmed after seeing their new home?	NO	Was the family relieved after seeing their new home?	YES
31	It was profoundly soul-stirring seeing the parents reduced to tears upon their arrival into their new homeland	Were the parents' tears soul-stirring?	YES	Were the parents' tears unmoving?	NO
32	It evoked a great deal of satisfaction when the family finally had their own beds to sleep in	Was it unusual to see the family sleep in their own beds?	NO	Was it satisfying to see the family sleep in their own beds?	YES
33	It was a welcome moment of peace and contentment watching the children bond with the other kids at the camp	Were the children bonding at the camp?	YES	Were the children keeping to themselves?	NO
34	There was a strong sense of warmth and fellowship when the family welcomed the stray dog into their arms	Did the family welcome a stray cat into their arms?	NO	Did the family welcome a stray dog into their arms?	YES
35	It instilled a sense of assurance in the goodness of humanity when the family comforted the distressed child	Was the family's help assuring?	YES	was the family's help unassuring?	NO
36	It induced a great deal of happiness to see watch the family form lasting bonds with new friends	Was it sad to see the family try to make new friends?	NO	Was watching the family try to make friends a happy sight?	YES
37	It was very relieving when the children found shelter to hide under during the severe storm	Did the children find cover to hide under during the storm?	YES	Did the children get caught out in the storm?	NO
38	It was touching to see the family's deep appreciation at having a roof to sleep under when in the refugee camp	Did the family sleep under the stars at the camp?	NO	Did the family sleep under a roof at the camp?	YES
39	It was very encouraging when, despite the brutal living conditions, the boat provided the family with shelter	Was it encouraging to see the boat provide shelter?	YES	Was it discouraging to see the boat provide shelter?	NO
40	It was an immensely joyous moment when the family finally had their own house and continuous shelter	Was it unremarkable to see the family in their own home?	NO	Was it a joyous moment to see the family in their own home?	YES
41	It was truly emancipating watching the family grow from a position of helplessness to one of happiness	Was the family's growth in a positive direction?	YES	Was the family's growth in a negative direction?	NO
42	It was extremely enriching to see the family progress towards a better life, despite the hardships they've endured	Was the family's journey without its hardships?	NO	Did the family encounter hardships in their journey?	YES
43	It was incredibly inspirational seeing how bravely the family sacrificed their home in search of a new one	Was the family's sacrifice a brave choice?	YES	Was the family's sacrifice cowardly?	NO

Table 2 (continued)

	Benign information statements	Benign information statement comprehension questions set 1	Correct answer	Benign information statement comprehension questions set 2	Correct answer
44	It was astonishing to witness the family build new friendships after having to leave their old friends	Was it difficult to watch the family make new friends?	NO	Was it astonishing to watch the family make new friends?	YES
45	It was awesome to see the children bond through their shared adversity, even though they used to fight	Did the children used to fight with eachother?	YES	Did the children always get along well?	NO
46	It evokded a sense of glee watching the family's unadulterated joy at being reunited after being separated	Was the family always together?	NO	Was the family separated?	YES
47	It was profoundly touching when the family realised how lucky they were to still have the support of eachother	Was the family's realisation touching?	YES	Was the family's realisation unmoving?	NO
48	It was a pleasing contrast to see how easily the family interacted with new people after their past adversities	Was it challenging to watch the family interact with new people?	NO	Was it pleasing to watch the family interact with new people?	YES
49	It was exceptional how, despite the awful living conditions, the family showed gratitude at being given shelter	Did the family show gratitude?	YES	Did the family forget to show gratitude?	NO
50	It really captured humanity's capacity for affection to see the family grow to appreciate one another	Did the family grow to resent one another?	NO	Did the family grow to appreciate one another?	YES
51	It elicited a uniquely raw sense of appreciation seeing the family learn to engage with life's simple joys	Did this moment elicit a sense of appreciation?	YES	Did this moment elicit a sense of unappreciation?	NO
52	It was energizing to see the kids' excitement in anticipation of their first meal in their new home	Were the kids' reactions heart-breaking?	NO	Were the kids' reactions energizing?	YES
53	It was a delight to witness the way the father kept his kids hopeful by describing how he invsioned their new life	Did the father have optimistic ambitions?	YES	Did the father have pessimistic ambitions?	NO
54	It was invigorating to hear how passionately the mother talked about her dream of owning a café one day	Did the mother dream of owning a bakery?	NO	Did the mother dream of owning a café?	YES
55	It was so lovely to hear the young boy talk about how he wanted to help other refugees in need	Was it lovely hearing the boy talk about his dreams?	YES	Was it depressing hearing the boy talk about his dreams?	NO
56	It was remarkably jovial listening to the children describe how they pictured their dream home	Was it discouraging listening to the kids describe their dream home?	NO	Was it jovial listening to the kids describe their dream home?	YES
57	There was a moment of pure exaltation when the father risked his own life to save the elderly man from drowning	Did the father risk his life to save the elderly man?	YES	Did the mother risk her life to save the elderly man?	NO

Table 2 (continued)

	Benign information statements	Benign information statement comprehension questions set 1	Correct answer	Benign information statement comprehension questions set 2	Correct answer
58	It was very comforting seeing the family's relief at having their belongings returned after the thug was stopped	Did the family lose their belongings?	NO	Did the family gain their belongings back?	YES
59	It was truly rousing to witness the bond that kept the family going despite them were being starving and tired	Was the family's bond a rousing sight?	YES	Was the family's bond a dispiriting sight?	NO
60	It was nothing short of spectacular to see the family's relief when the freak storm passed on by their village	Was it terrifying to witness the family's reaction?	NO	Was it spectacular to see the family's reaction?	YES
61	It was a welcome change when the family was granted passage for the boat when they couldn't afford tickets	Did the family make it onto the boat?	YES	Did the family get rejected from the boat?	NO
62	It was superb to see the care and diligence of the camp staff who quickly provided medicine for the sick child	Were the staff cold and unsympathetic?	NO	Were the staff caring and diligent?	YES
63	It was extremely heartening to see how hard the staff worked to make room for the family in the shelter	Was it heartening when the staff made room for the family?	YES	Was it disheartening when the staff made room for the family?	NO
64	It was a motivating gesture when the man donated a large sum of money to the family when they were begging	Was the gentleman's donation discouraging?	NO	Was the gentleman's donation motivating?	YES

Table 3 A list of Negative Information Statements and Negative Information Statement Comprehension Question pairs

	Negative information statements	Negative information statement comprehension questions set 1	Correct answer	Negative information statement comprehension questions set 2	Correct answer
1	When the father had to put down the badly injured horse, it was a confronting and horrific thing to watch	Was the horse badly injured?	YES	Was the horse uninjured?	NO
2	When the family had to leave all their friends behind and everyone was so upset, it was very disheartening	Was everyone excited for the family to leave?	NO	Was everyone upset that the family was leaving?	YES
3	Seeing the family abused, and treated as less than human, by these drunken soldiers was appalling	Was the family's treatment appalling?	YES	Was the family's treatment humane?	NO
4	When the father of the other family drowned, while trying to save his child, it was absolutely dreadful	Was it relieving watching the father try to save his child?	NO	Was it dreadful watching the father try to save the child?	YES
5	The images of the terrible injuries caused when the bomb destroyed the refugee centre were sickening	Did the bomb destroy the refugee centre?	YES	Did the bomb destroy the market?	NO
6	The fact that nobody helped when the daughter was crushed in the crowded refugee truck was utterly disgraceful	Was the youngest son crushed in the crowded truck?	NO	Was the youngest daughter crushed in the crowded truck?	YES
7	Seeing the beach filled with the bodies of so many entire families, after the boat accident, was highly traumatic	Was it traumatic seeing the beach?	YES	Was it heartwarming seeing the beach?	NO
8	When the parents were cheated by the trafficker, losing their entire life savings, it was extremely dispiriting	Was their interaction with the trafficker heartwarming?	NO	Was their interaction with the trafficker dispiriting?	YES
9	The intense distress of the daughter when she was brutally separated from her family was heart-wrenching	Was the daughter separated from her family?	YES	Was the son separated from his family?	NO
10	The abject squalor of the living conditions endured by the family at the camp was nauseatingly repugnant	Were the camp living conditions good?	NO	Were the camp living conditions bad?	YES
11	When the parents had to choose which of their children would live, and which would die, their agony was unbearable	Was it agonising to watch the parents make this choice?	YES	Was it easy to watch the parents make this choice?	NO
12	The image of that young boy taking his last breath, so skinny and weak from starvation, was unforgettable	Was it a relief to see the young boy take his last breath?	NO	Was it unforgettable to see the young boy take his last breath?	YES
13	When the family had to watch helplessly as the fire destroyed all their shared memories, it was terribly upsetting	Did the fire destroy their shared memories?	YES	Did the flood destroy their shared memories?	NO

Table 3 (continued)

	Negative information statements	Negative information statement comprehension questions set 1	Correct answer	Negative information statement comprehension questions set 2	Correct answer
14	The pleasure with which the pompous official denied the desperate family their entitled support was infuriating	Did the official give the family their entitled support?	NO	Did the official refuse the family their entitled support?	YES
15	The father's intense grief when he learned that the mother's ill-health was due to cancer was harrowing	Was the father's realisation harrowing to see?	YES	Was the father's realisation uplifting to see?	NO
16	The brutality of the father's severe beating at the hands of the port authority guards was shocking	Was the father's severe beating unsurprising?	NO	Was the father's severe beating shocking?	YES
17	The sight of the young boy being beaten in front of his father by thugs was quite simply revolting	Was the young boy beaten by thugs?	YES	Was the young girl beaten by thugs?	NO
18	Seeing the malicious guard leave the family's dog bloodied and broken in the streets was a nauseating sight	Did the guard help the dog?	NO	Did the guard leave the dog?	YES
19	The look of unadultered fear in the kids' eyes when having racial slurs shouted at them was enraging	Was it enraging to see how the children are treated?	YES	Was it encouraging to see how the children are treated?	NO
20	The hopelessness the family felt when angrily told to go back to their own country was disappointing	Was the family's reaction encouraging?	NO	Was the family's reaction disappointing?	YES
21	The way the ticket master yelled and screamed at the family for wasting his time was nothing short of outrageous	Did the ticket master yell at the family?	YES	Did the ticket master talk nicely to the family?	NO
22	The racism the family had to endure in their new homeland was extremely difficult to witness	Was everyone kind to the family in their new home?	NO	Did the family endure racism in their new home?	YES
23	Watching the family slowly become skinnier and more beaten-down over time was particularly difficult to endure	Was it difficult watching the family change over time?	YES	Was it easy watching the family change over time?	NO
24	To watch the parents' starve themselves so that their kids could eat each night was heartbreaking	Was it heart-warming to watch the parent's suffer for their kids?	NO	Was it heartbreaking to watch the parent's suffer for their kids?	YES
25	The sight of hundreds of refugees starving and exhausted at the camp was truly traumatising	Were there lots of refugees at the camp?	YES	Were there only a few refugees at the camp?	NO
26	The sight of rotting fish and waste aboard the refugee boat was an incredibly abhorrent sight	Was the refugee boat in a good condition?	NO	Was the refugee boat in a bad condition?	YES
27	Watching as the family desperately tried to eat the meat covered in filth and flies was unendurable	Was it unendurable to watch the family eat the meat?	YES	Was it relieving to watch the family eat the meat?	NO
28	The sight of the filth and mess that littered the makeshift refugee camp bathroom was truly horrifying	Was the state of the bathroom an uplifting sight?	NO	Was the state of the bathroom a horrifying sight?	YES

Table 3 (continued)

	Negative information statements	Negative information statement comprehension questions set 1	Correct answer	Negative information statement comprehension questions set 2	Correct answer
29	The sight of the sewage and waste that contaminated the family's home street was very haunting	Was the street lined with sewage and waste?	YES	Was the street clean?	NO
30	The fear in the father's eyes after being informed that he has tuberculosis was incredibly disheartening	Did the father learn that he has cancer?	NO	Did the father learn that he has tuberculosis?	YES
31	Seeing footage of sick refugees at the camp, desperately fighting the urge to give in to death, was gut-wrenching	Was the footage of the refugees gut-wrenching?	YES	Was the footage of the refugees inspiriting?	NO
32	The way the children cried and grieved when they discovered that their dog was fatally ill was tormenting	Was the childrens' grief a welcome sight?	NO	Was the children's grief tormenting?	YES
33	Watching as the mother screamed and cried when her husband was placed on a separate boat was crushing	Were the parents placed on separate boats?	YES	Were the parents placed on the same boat?	NO
34	The force with which the guard ripped the young girl from her mother's arms was extremely challenging to watch	Was the young boy ripped from his mother's arms?	NO	Was the young girl ripped from her mother's arms?	YES
35	The way the boy stared at the wall with vacant eyes after being torn from his family was especially painful to see	Was it painful watching the boy's reaction?	YES	Was it painless watching the boy's reaction?	NO
36	Watching the father grieve as his friend was executed in the street was unfathomably poignant to watch	Was the father's position a happy one?	NO	Was the father's position a sad one?	YES
37	When the boat wouldn't turn back for the man who fell overboard, the shrieks of terror that followed were scarring	Did a man fall from the boat?	YES	Did a woman fall from the boat?	NO
38	Watching the man having to endure burying his own son, his only surviving family, was incredibly heart-rending	Does the man still have surviving family?	NO	Is the man the last of his family?	YES
39	Watching the boy return home to find his stuffed bear charred and matted was very demoralising	Was the boy's return home demoralising?	YES	Was the boy's return home encouraging?	NO
40	Watching as the last family photo flew into the sky, carried by the wind, was quite dejecting	Was the sight of the family photo heartening?	NO	Was the sight of the family photo dejecting?	YES
41	The family's excitement quickly turned to despair when turned away by the coast guard, which was awful to see	Was the family rejected by the coast guard?	YES	Was the family accepted by the coast guard?	NO
42	The hostility the family endured at the hands of the citizens of their new home was truly deplorable	Were the citizens of their new home friendly?	NO	Were the citizens of their new home hostile?	YES

Table 3 (continued)

	Negative information statements	Negative information statement comprehension questions set 1	Correct answer	Negative information statement comprehension questions set 2	Correct answer
43	When the family was asked to leave the store because of how they were dressed, it was especially inflaming	Was the shop-keepers reaction inflaming?	YES	Was the shop-keepers reaction justified?	NO
44	Watching how hard it was for the family to find friends who accepted their tragic history was discouraging	Was it encouraging watching the family try to make friends?	NO	Was it discouraging watching the family try to make friends?	YES
45	To see the family take shelter in the ruins of their destroyed home, amid the ash and rain, was very depressing	Did the family find shelter at their home?	YES	Did the family fail to find shelter at their home?	NO
46	The fact that the family spent their nights wandering the streets looking for food scraps was particularly grim	Did the family sleep at night?	NO	Did the family wander the streets at night?	YES
47	Seeing so many families sleeping on the street, surrounded by dirt and disease, induced a sense of hopelessness	Did it feel hopeless seeing these families together?	YES	Did it feel hopeful seeing these families together?	NO
48	The sight of two young siblings trudging through the muck, begging for scraps of food or coin, was so tragic	Was the sight of the two siblings jolly?	NO	Was the sight of the two siblings tragic?	YES
49	Watching the family endure so much pain without anything to show for it induced an acute sense of despair	Was the family's pain in vain?	YES	Was the family's pain rewarded?	NO
50	The way the guard smirked when refusing the family entry to the boat, leaving them feeling hopeless, was maddening	Did the family feel hopeful?	NO	Did the family feel hopeless?	YES
51	Watching the father step outside and away from his family, only to break down in tears, was truly miserable	Was the father's faltering strength miserable to see?	YES	Was the father's faltering strength admirable to see?	NO
52	To watch the kids grow up with such suffering was a disturbing reminder of the inequity in this world	Was this reminder an inspiring one?	NO	Was this reminder a disturbing one?	YES
53	When the father put down their dog and tears rolled down his cheek, it was a particularly bereaving moment	Did the father struggle with this task?	YES	Did the father find this task easy?	NO
54	The family starved for days to afford boat tickets, which was a distressing contrast to the comfort we know	Could the family easily afford the boat tickets?	NO	Did the family acquire the boat tickets?	YES
55	For no good reason, the ticket vendor increased the cost of entry for the family, which was completely vile	Was the ticket vendor's action vile?	YES	Was the ticket vendor's action kind?	NO

Table 3 (continued)

	Negative information statements	Negative information statement comprehension questions set 1	Correct answer	Negative information statement comprehension questions set 2	Correct answer
56	Watching the guard threaten to toss the boy overboard if his parents didn't pay him was absolutely terrifying	Did the guard toss the boy overboard?	NO	Did the guard threaten to toss the boy overboard?	YES
57	The state of the refugee shelter, overcrowded and writhing with despair, was truly disgusting	Was the shelter overcrowded?	YES	Was the shelter roomy?	NO
58	The mother woke up one night to the touch of a rat crawling across her body, which was utterly nauseating	Did a spider crawl across the mother's body?	NO	Did a rat crawl across the mother's body?	YES
59	Watching as the family had to navigate through crowds of thugs and criminals just to return home was atrocious	Was the family's need for caution atrocious?	YES	Was the family's need for caution pleasing?	NO
60	Watching as the law enforcement exploited and abused the refugees was inexplicably gruelling	Was the law enforcement a positive presence?	NO	Was the law enforcement an abusive presence?	YES
61	The pictures of the massacre that occurred in one of the smaller villages were truly mortifying to behold	Did the massacre occur in a village?	YES	Did the massacre occur in the desert?	NO
62	The silence that followed the execution of a boy as punishment for stealing an apple was unsettling	Was the boy's punishment fair?	NO	Was the boy's punishment unjust?	YES
63	The fear lingering in the eyes of the boy's parents after his wound nearly took his life was perturbing	Were the parents' reactions perturbing?	YES	Were the parents' reactions relieving?	NO
64	The lack of empathy shown by the guard when asked to stop the thugs from beating an innocent man was contemptible	Was the guard's attitude helpful?	NO	Was the guard's attitude unhelpful?	YES

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Research Involving Human and Animal Participants No animal studies were carried out by the authors for this article.

Informed Consent All procedures followed were in accordance with the ethical standards of the responsible committee on human experimentation (national and institutional). Informed consent was obtained from all individual subjects participating in the study.

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